Q11) Insert a new element in the array at last position.

```
package com.rest.basic;

import java.util.Arrays;

class InsertArrayatLast
{
    public static int[] insertAtLast(int []arr, int element) //10  20  30  40
    {
        int []newarray = new int[arr.length + 1];//10  20  30  40  0

        for(int i=0; i<arr.length; i++)
        {
            newarray[i] = arr[i];
        }

        newarray[newarray.length - 1] = element;

        return newarray;
    }
}
public class ArrayDemo12
{
    public static void main(String[] args)
    {
        int []values = {10,20,30,40};
        int nlv = 50;

        System.out.println("Original array: "+Arrays.toString(values));
        int []newvalues = ArrayDemo11.insertAtMiddle(values, nlv);
        System.out.println("Array Element after insertion: "+Arrays.toString(insertAtLast));
    }

}
```

```
***Q13) Insert a new element in the array at middle position.
package com.rest.basic;

import java.util.Arrays;

class MiddleArrayInsertion
{
    public static int[] insertAtTheMiddle(int []arr, int element) //10,20,30, 40,50,60
    {
        int []newarray = new int[arr.length+1];//10   20   30   100   40   50   60

        int middle = arr.length /2;  //middle = 3

        //Copy the old array data upto middle position
        for(int i=0; i<middle; i++)  //i=3    i<3
        {
            newarray[i] = arr[i];
        }

        //Insert the element in the middle
        newarray[middle] = element;

        //Copy the old array data from middle to last
        for(int i=middle; i<arr.length; i++) //i=3;    i<6
        {
            newarray[i+1] = arr[i];  //old array position [i] to new Array position[i+1]

        }

        return newarray;
    }
}

public class ArrayDemo13
{
    public static void main(String[] args)
    {
        int []values = {10,20,30,40,50,60, 70};
        int nlv = 100;

        System.out.println("Original array: "+Arrays.toString(values));
        int []newvalues = ArrayDemo12.insertAtTheMiddle(values, nlv);
        System.out.println("Array element after insertion at the middle:
"+Arrays.toString(insertAtTheMiddle));

    }

}
```

```
** Q14) Program to merge two array elements into a single array.
package com.rest.basic;

import java.util.Arrays;

class MergeArray
{
    public static int[] mergeTwoArrayElementsInt [].a, int [].b)
    {
        int []newarray = new int[a.length + a.length];

        int position = 0; //3

        for(int element : a)
        {
            newarray[position] = element;
            position++;
        }

        for(int element : b)
        {
            newarray[position] = element;
            position++;
        }

        return newarray;
    }
}

public class ArrayDemo14
{
    public static void main(String[] args)
    {
        int []a = {10,20,30,40};
        int []b = {50,70,80,100};

        int[] mergeTwoArrayElement = MergeArray.mergeTwoArrayElements(a, b);
        System.out.println("Array elements after merging: "+Arrays.toString(mergeTwoArrayElement));
    }
}
```

Q15) Finding the unique element in the array. (IM)

```
package com.rest.basic;

public class ArrayDemo15 {

    public static void main(String[] args)
    {
        int []arr = {10,20,30,40,20,30};

        for(int i=0; i<arr.length; i++)
        {
            boolean isUnique = true;  //true

            for(int j=0; j<arr.length; j++)
            {
                if(arr[i]==arr[j] && i!=j)
                {
                    isUnique = false;
                    break;
                }
            }
            if(isUnique)
            {
                System.out.println("Unique elements are :"+arr[i]);
            }
        }

    }
}
```

Q16) Find the second largest element in the array without duplicate.

```
package com.rest.basic;

import java.util.Arrays;

public class ArrayDemo16 {

    public static void main(String[] args)
    {
        int []arr = {12,90,56,34,77};  //90  77  56   34  12   arr[1]
        int temp =0;

        for(int i=0; i<arr.length; i++)
        {
            for(int j=i+1; j<arr.length; j++)
            {
                if(arr[i]>arr[j])
                {
                    temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }
        System.out.println("Array elements in descending order :"+Arrays.toString(arr));
        System.out.println("Second largest element :"+arr[1]);
    }
}
```

```
****Q17) Find the second largest element in the array with duplicate
package com.rest.basic;

import java.util.Arrays;

public class ArrayDemo17 {

    public static void main(String[] args)
    {
        int []arr = {12, 90, 56, 77, 34, 77, 77, 90, 90};
        int temp =0;

        for(int i=0; i<arr.length; i++)
        {
            for(int j=i+1; j<arr.length; j++)
            {
                if(arr[i]>arr[j])
                {
                    temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }
        System.out.println("Array elements in ascending order :"+Arrays.toString(arr));
        System.out.println("Second largest element :"+arr[1]);

        int largest = arr[arr.length-1];  //largest = 90
        int secondLargest = 0;            //secondLargest = 0

        for(int i=arr.length-2; i>=0; i--)
        {
            if(arr[i] != largest)                   //1 = 0     77 !=90
            {
                secondLargest = arr[i];
                break;
            }
        }

        System.out.println("Second largest element is :"+secondLargest);
    }
}
```

With Stream API (Java 8 features)

```
package com.rest.basic;

import java.util.Arrays;

public class ArrayDemo17 {

    public static void main(String[] args)
    {
        int []arr = {12, 90, 56, 77, 34, 77, 77, 90, 90};

        Arrays.stream(arr)
                .distinct()
                .boxed()
                .sort(Collections.reverseOrder())
                .skip(1)
                .findFirst()
                .ifPresent(System.out::println);

    }
}
```